Bus-Sense: Smart Bus Management System

Abstract

Bus-Sense is an intelligent smart bus monitoring system designed to improve public transportation efficiency using computer vision and IoT technologies. AI-enabled cameras capture video frames inside buses and a CNN model detects passengers and calculates occupancy levels. GPS modules track bus location and send real-time information to a cloud database. A web dashboard displays occupancy, location, and analytics to transport authorities and passengers. The system improves safety, prevents overcrowding, and enables data-driven transport management.

Introduction

Urban transportation systems face overcrowding, lack of monitoring, and inefficient route planning. Smart technologies like AI, IoT, and cloud computing can solve these challenges by enabling automated passenger detection and real-time tracking.

Problem Statement

Public transportation systems lack real-time passenger monitoring and accurate occupancy data. This results in overcrowded buses and inefficient route management.

Objectives

Develop an AI-based passenger detection system, track bus location using GPS, provide real-time dashboards, and support analytics for transport optimization.

System Architecture

The system consists of cameras, CNN detection modules, GPS trackers, a cloud database, and a web dashboard.

Implementation

Camera frames are processed by CNN models. Passenger counts and GPS data are sent to the cloud database. The web dashboard displays the data in real time.

Database Design

Tables include Bus, Passenger_Count, and GPS_Location storing bus IDs, passenger counts, and location data.

Results

The system accurately estimates passenger occupancy and provides real-time monitoring.

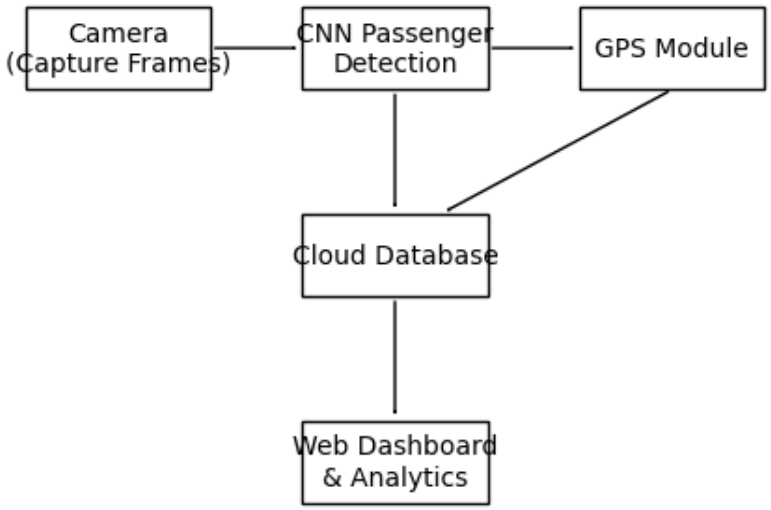
Future Work

Future improvements include mobile apps, predictive analytics, and smart ticketing integration.

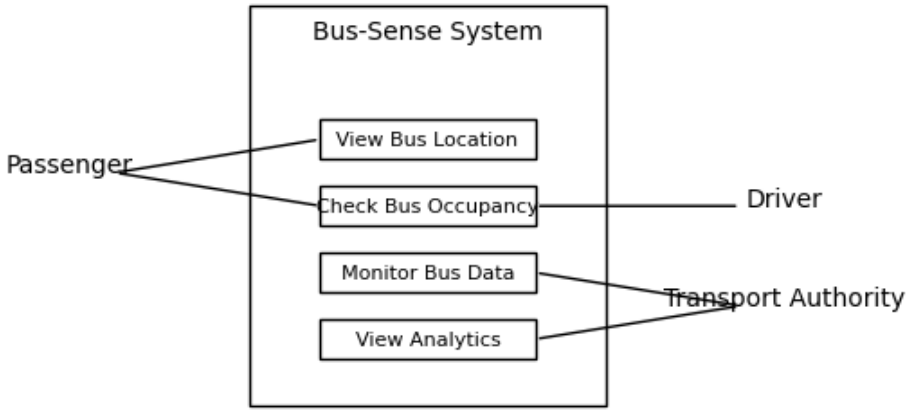
Conclusion

Bus-Sense enhances public transport management through AI-powered passenger detection and real-time monitoring.

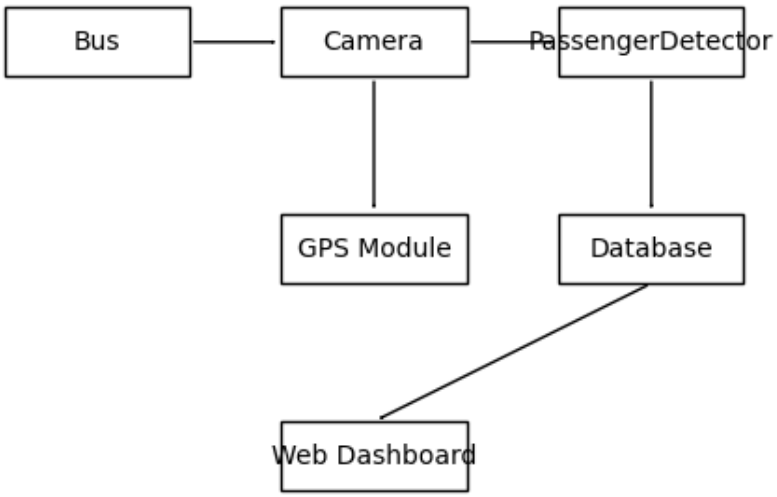
System Architecture Diagram



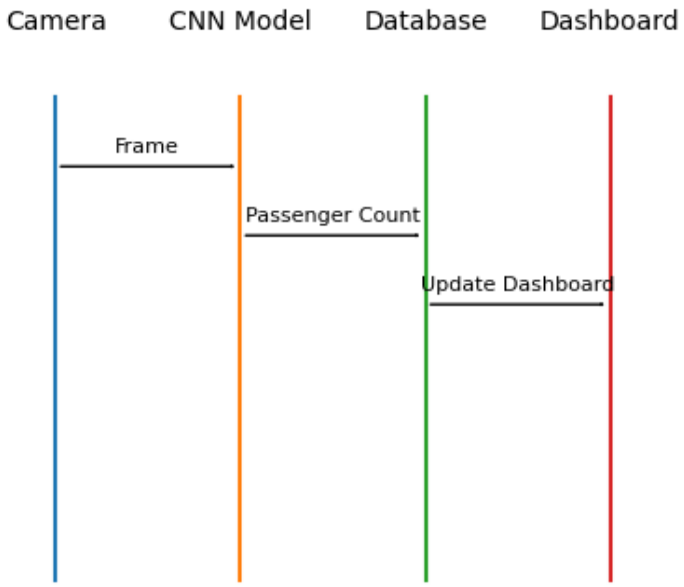
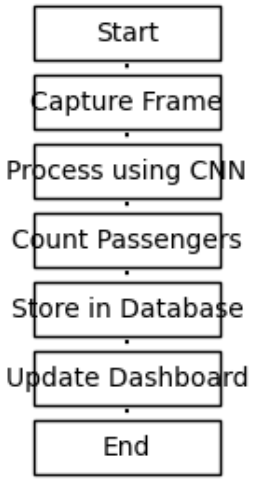
Use Case Diagram



Class Diagram



Sequence Diagram



Activity Diagram